

Breast-conserving surgery for breast cancer

Breast-conserving surgery aims to remove the cancer safely, but keep as much of the breast tissue and breast shape as possible.

On this page

- [What is breast-conserving surgery?](#)
- [Removing a larger area of breast tissue](#)
- [Radiotherapy after breast-conserving surgery](#)
- [Clear margins](#)
- [About our information](#)
- [How we can help](#)

What is breast-conserving surgery?

Breast-conserving surgery removes the cancer while keeping as much of the breast tissue and shape as possible. This type of surgery is called a wide local excision (WLE). You might also hear it called a lumpectomy.

During a WLE operation, the surgeon removes the cancer and some normal-looking tissue around it. This is called the margin.

Before the operation, a doctor inserts a fine wire or marker through the skin into the area where the cancer is. This is called [wire localisation](#). The wire or marker is secured to your chest with tape or a dressing. The doctor uses an ultrasound or x-ray to help place the marker so that it marks the area to be removed. When the surgeon does the operation, they can also use x-ray or ultrasound to help them find the right area more easily. During the operation, the surgeon removes the wire, the cancer and some surrounding tissue.

You may have a magnetic seed injected into the area instead of having wire localisation. The surgeon uses a handheld machine that can detect the seed and the area to be removed. Your cancer doctor or breast care nurse can explain more about magnetic seeds.

If a large amount of tissue is removed, the breast will be smaller than before. If this happens, the surgeon can reduce the size of your other breast. This can help make your breasts look a similar size.

Removing a larger area of breast tissue

Depending on the size of the cancer, you may need to have a larger area of breast tissue removed.

In this situation, surgeons can use different ways to help improve the appearance of your breast after the operation. They may reshape the breast by moving the breast tissue around and making the breast smaller. Sometimes they take tissue from somewhere else in the body to help reshape the breast. This is called breast reconstruction.

Your surgeon may suggest you have the other breast made smaller so that both breasts look a similar size. This can be done at the same time as your operation or in a separate operation later.

We have more information about [breast reconstruction](#) for if you are having breast-conserving surgery and surgery to reshape the breast.

Radiotherapy after breast-conserving surgery

Your surgeon will usually advise you to have [radiotherapy](#) after a WLE. This reduces the risk of the cancer coming back in the breast.

Having breast-conserving surgery and radiotherapy is usually as effective as having a mastectomy.

Clear margins

After breast-conserving surgery, a pathologist looks at the tissue that has been removed under a microscope. They check the area around the cancer. This is called the margin. You will need another operation to remove more tissue if:

- there is [ductal carcinoma in situ \(DCIS\)](#) or some types of [lobular carcinoma in situ \(LCIS\)](#) close to the edge of the area
- there are any cancer cells close to the edge of the area.

If the margins are clear, this reduces the risk of cancer coming back in the breast.

If your surgeon does not think another breast-conserving operation is likely to be successful, they may recommend a mastectomy. In this situation, you will usually be offered breast reconstruction.

About our information

References



Below is a sample of the sources used in our breast cancer information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

ESMO. Early breast cancer clinical practice guidelines for diagnosis, treatment and follow-up.

2019, Vol 30, pp1192–1220. Available from: <https://www.esmo.org/guidelines/guidelines-by-topic/breast-cancer/early-breast-cancer> [accessed 2023].

National Institute for Health and Care Excellence (NICE). Early and locally advanced breast cancer: diagnosis and management. 2018. Updated 2023. Available from: <https://www.nice.org.uk/guidance/ng101> [accessed 2023].

Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by [Dr Rebecca Roylance](#), Consultant Medical Oncologist and [Professor Mike Dixon](#), Professor of Surgery and Consultant Breast Surgeon.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

Reviewed: 01 October 2023 | Next review: 01 October 2026

ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help